FlashAttention changes the movement order, not the attention formula

HBM (off-chip)
tens of GB, a few TB/s
all T positions of K, V live here

load tile j

$j = 1, \ldots, T/M$

on-chip shared memory
$\sim$100 kB per compute block
far higher bandwidth
Q_i K_j V_j

O written once
$T \times d_v$

online-softmax state — updated once per K/V tile

$S_{ij} = Q_i K_j^\top / \sqrt{d_k}$ local scores, never leave the chip

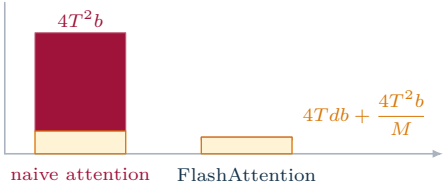
$m' = \max(m, \text{rowmax}(S_{ij})), \quad r = e^{m-m'}$

$\ell' = r\ell + \text{rowsum}(e^{S_{ij}-m'})$ running sum

$O' = (r\ell O + e^{S_{ij}-m'} V_j)/\ell'$ partial output

commit $(m, \ell, O) \leftarrow (m', \ell', O')$; rescale old sums by r

HBM traffic per attention layer



no $T \times T$ tensor in HBM

FLOPs unchanged; bytes moved changed

peak memory $O(T^2) \rightarrow O(T)$

rose: materialising the $T \times T$ score tensor; amber: the unavoidable $O(Td)$ reads of Q, K, V